CHAPTER 1

REFINING THE BASIC NEEDS APPROACH: A MULTIDIMENSIONAL ANALYSIS OF POVERTY IN LATIN AMERICA

Maria Emma Santos, Maria Ana Lugo, Luis Felipe Lopez-Calva, Guillermo Cruces and Diego Battiston

ABSTRACT

Latin America has a longstanding tradition in multidimensional poverty measurement through the unsatisfied basic needs (UBN) approach. However, the method has been criticized on several grounds, including the selection of indicators, the implicit weighting scheme and the aggregation methodology, among others. The estimates by the UBN approach have traditionally been complemented (or replaced) with income poverty estimates. Under the premise that poverty is inherently multidimensional, in this chapter we propose three methodological refinements to the UBN approach. Using the proposed methodology we provide a set of comparable poverty estimates for six Latin American countries between 1992 and 2006.

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1. INTRODUCTION

Poverty is being increasingly recognized as an inherently multidimensional phenomenon. Welfarists stress both the existence of market imperfections and incompleteness and the lack of perfect correlation between relevant dimensions of well-being (Atkinson, 2003; Bourguignon & Chakravarty, 2003; Duclos & Araar, 2006). Nonwelfarists point to the need to move away from the space of utilities to a different and usually wider space, where multiple dimensions are both instrumentally and intrinsically important.

Among the nonwelfarists, there are two main strands: the basic needs approach and the capability approach (Duclos & Araar, 2006). The first approach focuses on a set of primary goods that are constituent elements of well-being and considered necessary to live a good life. "Basic needs may be interpreted in terms of minimum specified quantities of such things as food, clothing, shelter, water and sanitation that are necessary to prevent ill health, undernourishment, and the like" (Streeten, Burki, Haq, Hicks, & Stewart, 1981, p. 25). "A basic needs approach to development (…) tries to ensure access to particular resources (such as caloric adequacy) for particular groups (defined by age, sex, or activity) that are deficient in these resources" (Streeten et al., 1981, pp. 33–34).

The second approach, introduced by Sen (1992, 2009), argues that attention should be shifted from the *means* of living to the *actual opportunities* a person has, and therefore argues that the relevant space of well-being should be the set of capabilities. These are defined as "the various combinations of functionings (beings and doings) that the person can achieve. Capability is, thus, a set of vectors of functionings, reflecting the person's freedom to lead one type of life or another … to choose from possible livings" (Sen, 1992, p. 40). The actual implementation of the capability approach poses some difficulties since one would need to observe the whole set of potential functionings. A "second best" alternative to implement this approach is to consider the actual functionings people have and evaluate poverty according to that. This presents a strong difference with the basic needs approach in that it requires evaluating the functioning and not the access to the resource that could eventually permit such functioning. For example, it would require looking into the actual nutritional status of each household member, and not the food consumption level, evaluating their cognitive skills and not whether they have had access to schooling. We understand that this is indeed the desirable approach to follow and we claim that surveys should be redesigned in order to capture actual functionings rather than access to resources. Even more, they should eventually aim at measuring capability sets. However, so

far, the availability of data poses serious constraints to what can be done, and that has probably been the main reason for the widespread use of the basic needs approach over the capability approach. Moreover, very frequently, and as it is the case of this chapter, the basic needs approach can be seen as a (very) imperfect proxy of the capability approach.

In this context, and until we can count with better sources of information that allow identifying actual functionings, we propose three specific refinements to the basic needs approach and we empirically implement them in six Latin American countries, Argentina, Brazil, Chile, el Salvador, Mexico, and Uruguay, over a period of 14 years (1992–2006).

2. THE BASIC NEEDS AND THE INCOME APPROACH TO POVERTY MEASUREMENT IN THE REGION

Latin America has a longstanding tradition in multidimensional poverty measurement making use of the basic needs approach. Promoted in the region by the United Nation's Economic Commission for Latin America and the Caribbean (ECLAC), the approach was employed extensively since the beginning of the 1980s (Feres & Mancero, 2001). In a context where household surveys were not as widespread as nowadays and income and consumption were difficult variables to measure, the census-based unsatisfied basic needs (UBN) measures became the poverty analysis tool of widespread use in the region, while income poverty studies were restricted to specific surveys and individual studies.[1]

Most commonly, the UBN approach combines population census information on the condition of households (construction material and number of people per room), access to sanitary services, children attending school and education, and economic capacity of household members (generally the household head). The UBN indicators are often reported by administrative areas in terms of the proportion of households unable to satisfy one, two, three, or more basic needs, and are often presented using poverty maps (Feres & Mancero, 2001). Thus, in practice, the approach does not offer a unique index but rather the percentage of population with different number of unmet basic needs.

As household surveys started to be regularly administered and progressively available to the public, distributional studies using income became widespread as well as official income poverty estimates started to be reported periodically. Since then both methods have coexisted.

The UBN method has also been called the "direct method" to measure poverty since it looks directly at whether certain needs are met or not, as opposed to the "indirect (or poverty line) method," which looks at the income level and compares it to the income level necessary to achieve these needs (Feres & Mancero, 2001). As argued in the Introduction, in practice, the basic needs approach considers access to resources (*means* rather than actual *ends*), so the reference as a *direct* method is arguable. It is however true that it looks at a wider range of resources than the income approach, and in particular it incorporates access to services such as education, sanitation, and water. Given the existence of imperfect markets, capturing access to basic services represents a value-added over the income approach.

Consistent with a multidimensional understanding of poverty, it has been long argued that both methods capture partial aspects of poverty, that both income as well as the UBN indicators are relevant for assessing well-being, and that there are significant errors in targeting the poor (either of inclusion or exclusion) when only one of them is used.[2] Therefore, one of the contributions of this chapter is to expand the UBN approach into a *hybrid* one, incorporating income. The reasons for doing so are that income (a) can act as a surrogate for all the other nonconsidered dimensions due to data restrictions,[3] (b) it has been found to have relatively low correlations with the other considered indicators,[4] (c) even when merely a means, having purchasing power provides the household with some freedom to choose the bundle of goods. We think that having a set of relevant indicators combined into one single measure can prove helpful for monitoring poverty and for policy design.

In terms of the UBN indicators themselves, there have been critiques arguing the selection to be arbitrary. Clearly in any multidimensional poverty measure (and in any composite indicator in general), the selection of indicators will be problematic, and some selection criteria are preferable to others from a methodological point of view (see Alkire, 2008 on such different methods). While the selection of indicators seems to have been originally highly influenced by data availability, it gained some form of public consensus over the years as estimates were periodically released. Drawing on such gained consensus, we use similar indicators to those of the UBN (with some minor adjustments explained in Section 2.2). However, we understand that this set of indicators is indeed very limited. We therefore advocate for an improvement in the collection of survey data that allows capturing other key variables today completely absent (such as health), as well as progressing toward the measurement of people's functionings. We also consider, as suggested by the capability approach and especially by

Sen (2009), that it would be important to reexamine and evaluate through public reasoning the dimensions to be considered in a multidimensional poverty index for the region. This could guide the subsequent improvement in data collection.

The two other methodological contributions of the chapter derive from the measure used. On the one hand, by using the first measure of the family of multidimensional poverty measures proposed by Alkire and Foster (2007), one can account not only for the poverty *incidence* (the percentage of people that are multidimensionally poor) but also for the *breadth* of poverty, that is, on average, the proportion of the considered indicators in which the poor are deprived. This represents a strong advantage over the usual multidimensional headcounts reported by the UBN approach. In addition, the measure used allows for alternative weighting systems which gives flexibility to the index, providing another strong advantage over the traditional UBN headcounts which had an implicit weighting system sometimes criticized.

In summary, the chapter proposes three specific refinements to the UBN methodology: (1) incorporating income as a proxy indicator for other nonincluded dimensions, (2) incorporating the breadth of poverty, (3) allowing for a flexible weighting system. Last but not least, the chapter contributes with empirical evidence on multidimensional poverty in the region. Six countries are covered with five observations over a period of 14 years (1992–2006). The countries under analysis are Argentina, Brazil, Chile, El Salvador, Mexico, and Uruguay, which altogether cover about 64 percent of the total population in Latin America in 2006.

The existing studies in this specific area are limited in the region. Amarante, Arim, and Vigorito (2008) and Arim and Vigorito (2007) study the case of Uruguay. The first compares three alternative methodologies; namely, Bourguignon and Chakravarty (2003) indices, the fuzzy sets approach, and the stochastic dominance approach developed by Duclos, Sahn, and Younger (2006). The second one only uses Bourguignon and Chakravarty (2003) family of indices. They find that multidimensional poverty has decreased and that its evolution over time is smoother than that of the income poverty, as the first one includes less volatile indicators. Conconi and Ham (2007) also employ Bourguignon and Chakravarty indices (but using a relative approach to measurement) in a study on Argentina for the period around the last financial crisis (1998–2002). The authors find that the increased deprivation in employment and income is behind the rising trend in poverty in the study period.

A number of other studies propose alternative measures of multidimensional poverty to study Latin American countries. Paes de Barros, De

Carvalho, and Franco (2006) suggest using a weighted average of dichotomous indicators of deprivations as a multidimensional poverty measure for Brazil. Ballon and Krishnakumar (2008) develop a multidimensional capability deprivation index based on structural equation modeling and apply the method to a household survey dataset for Bolivia in 2002, focusing on two children capability domains: knowledge and living conditions. The authors find a strong interdependence between the two studied dimensions. López-Calva and Rodríguez-Chamussy (2005) and López-Calva and Ortiz-Juárez (2009) have also adopted a multidimensional approach to studying poverty in Mexico. They estimate the magnitude of the "exclusion error" in targeting programes when a monetary measure is adopted instead of a multidimensional one. They find a large variability in the exclusion error depending on the selected criterion to identify the multidimensionally poor (union vs. intersection, explained in the next section). Finally, since 2004, the Programa Observatorio de la Deuda Social Argentina (Pontificia Universidad Catolica Argentina) implements a survey that collects information on housing conditions, health and subsistence, and computes a composite indicator of deprivation constructed using principal components analysis.

The results of the present chapter are quite encouraging: a decreasing trend in multidimensional poverty is observed, both in incidence and breadth of poverty. However, strong disparities between countries as well as between urban and rural areas within countries remain, which demand renewed efforts to reduce poverty in the region.

3. METHODOLOGY

3.1. Multidimensional Poverty Measurement: From H *to* M_0

In this section, we describe the measures used by the UBN approach and present a simple way in which the UBN index could be improved by using one of the members of the family of multidimensional poverty measures proposed by Alkire and Foster (2007).

In the multidimensional context, distributional data are presented in the form of a matrix of size $n \times d$, $\mathbf{X}^{n,d}$, in which the typical element x_{ij} corresponds to the achievement of individual i in dimension j, with $i = 1, \ldots, n$ and $j = 1, \ldots, d$. Vector $\boldsymbol{x}_i$ contains the achievements of individual i in the d dimensions. Analogous to the unidimensional approach, the measurement of poverty in the multidimensional approach involves two

steps (Sen, 1976): first the identification of the poor, second, the aggregation of the poor.

The most common approach for identifying the poor in the multidimensional context is to first define a threshold level for each dimension j, below which a person is considered to be deprived. The collection of these thresholds can be expressed in a vector of poverty lines $\boldsymbol{z} = (z_1, \ldots, z_d)$. In this way, whether a person is deprived or not in each dimension can be defined. However, unlike unidimensional measurement, a second decision needs to be made: among those who fall short in some dimension, who is to be considered multidimensionally poor? A natural starting point is to consider all those deprived in at least one dimension, the so-called *union approach*. Other more demanding criteria can be used, even to the extreme of requiring deprivation in all considered dimensions, the so-called *intersection approach*.

In terms of Alkire and Foster (2007), the number of dimensions in which someone is required to be deprived so as to be identified as multidimensionally poor constitutes a second cut-off (the first cut-off were the dimension-specific ones contained in vector $\boldsymbol{z}$). The authors name this second cut-off k and define c_i to be the number of deprivations suffered by individual i. Then, an identification function $\rho_k(x_i; z)$ is defined, such that:

$$\rho_k(x_i; z) = \begin{cases} 1 & \text{if } c_i \geq k \\ 0 & \text{if } c_i < k \end{cases} \tag{1}$$

In other words, if $c_i \geq k$, the individual is identified as multidimensionally poor, and if $c_i < k$, she is not, despite she may be experiencing some deprivation.

For the aggregation step, one natural first measure is the headcount ratio, also frequently known as the poverty incidence, which is the fraction of the population identified as being multidimensionally poor. It is simply given by:

$$H = \frac{\sum_{i=1}^{n} \rho_k(x_i, z)}{n} = \frac{q}{n} \tag{2}$$

where q is the number of people identified as multidimensionally poor. Clearly, the value of H varies with the selected k cut-off, decreasing as k increases. The H measure is what the UBN approach has used, most frequently (but not always) using a k cut-off of one, that is, the union approach.

Since the pioneer papers of Watts (1969) and Sen (1976), the limitations of the headcount ratio in the unidimensional approach have been repeatedly remarked, namely, that it is insensitive to the depth and distribution of poverty. In formal terms, it violates the monotonicity and transfer axioms.[5]

Those critiques also apply to the multidimensional headcount, and therefore, to the UBN approach. Moreover, as noted by Alkire and Foster (2007), given a k value, if an individual identified as poor becomes deprived in an additional dimension, the multidimensional headcount does not change, that is, it violates what the authors call *dimensional monotonicity*. In simpler words, it is insensitive to the *breadth* of poverty: the number of deprivations suffered by the poor.

Related to the latter point, Alkire and Foster (2007) argue that another informative measure is the average deprivation share across the poor, that is, the average fraction of dimensions in which the poor are deprived. This can be expressed as:

$$A = \frac{\sum_{i=1}^{n} c_i}{qd} \tag{3}$$

H and A can be easily combined into one single measure, called by the authors as M_0, which is just the headcount ratio "adjusted" (i.e., multiplied) by the breadth of poverty:

$$M_0 = HA \tag{4}$$

M_0, also called the adjusted headcount ratio, is the first member of the M_α family of measures proposed by the authors. This family constitutes an extension of the unidimensional Foster, Greer, and Thorbecke (1984) (FGT or P_α) measures to the multidimensional context.[6] Clearly, M_0 is sensitive to the breadth of poverty, that is, it satisfies dimensional monotonicity. If someone becomes poor in one additional dimension, A will increase and therefore M_0 will increase. Another example of the importance of the property is the following. Suppose two regions A and B, both with 50 percent of their population experiencing two or more deprivations. If in A that 50 percent experiences, on average, two deprivations out of six, while in B that 50 percent experiences on average four out of six, M_0 will be higher in B than in A.

It must be noted though that, as H, M_0 is not sensitive to the depth of poverty, which has been a usual critique of the UBN approach. If someone becomes more deprived in one dimension, M_0 will not change. The measure is also insensitive to the distribution of achievements among the poor. If one wants to account for the depth of poverty and for the distribution, other members of the M_α family need to be used (see Footnote 7). Alternatively, members of other proposed families of multidimensional poverty measures could be used, such as those introduced by Bourguignon and Chakravarty (2003), Tsui (2002), and Maasoumi and Lugo (2008). However, to incorporate considerations of depth and distribution with any of these

measures one would need cardinality in *all* the considered indicators. Unfortunately, this is not the case of many of the indicators usually considered under a multidimensional approach.[7] In summary, the failure of incorporating poverty depth in multidimensional measurement is not a consequence of the unavailability of appropriate aggregation methodologies, but of the nature of the indicators used. Given these constraints, the possibility that the M_0 measures offers of at least accounting for the breadth of poverty constitutes an important advantage over H and, therefore, over the UBN aggregation methodology.

Akin to their unidimensional counterpart, all the M_α measures can be decomposed by population subgroups, so that one can identify, for example, which are the regions that are contributing more to aggregate poverty.[8] Moreover, once the poor have been identified, the measures can be broken-down by dimension, so it is possible to determine to which extent the deprivation in each dimension contributes to overall multidimensional poverty.[9] This is a second advantage of M_0 over H.

There is a third advantage of the M_0 measure over H (which is also present in the other M_α measures): it allows for alternative weighting systems of the dimensions, which affect both the identification and the aggregation steps. So far, we have implicitly assumed equal weights for all the considered dimensions ($w_j = 1 \quad \text{for all} \quad j = 1, \ldots, d$). In that case, the identification cut-off ranges from $k = 1$, corresponding to the union approach, to $k = d$, corresponding to the intersection approach, and someone is multidimensionally poor when her number of deprivations is equal or greater than k: $c_i \geq k$. However, one could use a "ranking weighting system" so that some dimensions receive higher weights than others. The sum of the weights needs to equal the total number of dimensions considered d. In that case, c_i becomes the *weighted* number of deprivations in which the individual is deprived. For example, if an individual is deprived in income and health, and income has a weight of 2, while health has a weight of 0.5, then $c_i = 2.5$ and not 2, as it would be with equal weights. In this case, the minimum possible k value, which corresponds to the union approach, is given by the minimum weight: $k = \min(w_j)$, while the maximum possible k cut-off value remains to be d. With the mentioned change in the definition of c_i, the definition of H, A, and consequently M_0 are automatically adjusted to incorporate these weights.[10]

So far, we have referred to considering d dimensions, implicitly assuming that there is one indicator per dimension. However, in the practice of measurement, very often one considers more than one indicator referred to the same dimension. This is related to one critique sometimes done to the UBN approach: if there is more than one indicator corresponding to the

same dimension, this means that some dimensions are weighted disproportionately more than others (Feres & Mancero, 2001). In those cases, the possibility of alternative weighting systems is an important feature of the M_0 measure, which permits to overcome such critique. Indeed, one can use "nested weights." For example, if there are five indicators, three of which correspond to the same dimension, say for example, housing, while the other two correspond to different dimensions, say income and health, then one can assign a weight of 5/3 to income and health each, and a value of 5/9 to each of the three housing indicators, assuring that each of the three considered dimensions are equally weighted. Other schemes can also be used, as considered appropriate by the nature of the measurement exercise.

3.2. Data, Indicators, and Poverty Lines

The dataset used in the chapter corresponds to the *Socioeconomic Database for Latin America and the Caribbean* (SEDLAC), constructed by the Centro de Estudios Distributivos Laborales y Sociales (CEDLAS) and the World Bank. The dataset comprises household surveys of different Latin American countries that have been homogenized to make variables comparable across countries – the details of this process are covered in CEDLAS (2009). This study concentrates on a subset of the available database to maximize the possibilities for comparison across time and between countries.[11] The study covers Argentina, Brazil, Chile and Uruguay, El Salvador and Mexico. Altogether, they account for about 64 percent of the total population in Latin America in 2006.

The chapter performs estimates at five points in time between 1992 and 2006 for each country. Full details of survey names and sample sizes can be found in Table A1. In the case of Argentina and Uruguay, the data are representative only of urban areas and correspond to the years 1992, 1995, 2000, 2003, and 2006 in Argentina, and to the years 1992, 1995, 2000, 2003, and 2005 in Uruguay. In the other four countries data are nationally representative, including information from both urban and rural areas. In Brazil, data corresponds to the years 1992, 1995, 2001, 2003, and 2006; in Chile to 1992, 1996, 2000, 2003, and 2006; in El Salvador to 1991, 1995, 2000, 2003, and 2005; and finally in Mexico, to the years 1992, 1996, 2000, 2004, and 2006. The definition of "rural areas" by the surveys performed in each of these four countries is fairly similar.[12] In each country, only households with complete information on all variables and consistent answers on income were considered.[13]

Table 1. Selected Indicators and Deprivation Cut-Off Values.

Dimension	Indicator	Deprivation Cut-Off Value
Command over resources	Income	Having a per capita family income of US$ 2.
Education	Child in school	Having all children between 7 and 15 years old attending school.
	Education of HH	Household head with at least five years of education.
Housing	Running water	Having tap water in the dwelling.
	Sanitation	Having flush toilet in the dwelling.
	Shelter	House with nonprecarious wall materials.

Table 1 presents the dimensions and indicators selected to perform the poverty estimates. The selected indicators can be seen as pertaining to three different dimensions (although other groupings are also possible): command over resources, education of the household, and housing conditions. For the income indicator (command over resources), the World Bank's poverty line of US$2 per capita per day is used. It is acknowledged that this is a rather conservative poverty line for Latin America, but it guarantees full comparability across countries.[14,15] Education of the household contains two indicators. One is whether children between 7 and 15 years old (inclusive) are attending school. This indicator belongs to the UBN approach. Households with no children are considered nondeprived in this indicator. The other education indicator refers to the educational level of the household head, with the threshold set at five years of education. Again this indicator is part of the UBN approach, although in that approach (a) the required threshold is second grade of primary school and (b) it is usually part of a composite indicator of "subsistence capacity" together with the dependency index of the household (considered to be deprived if there are four or more people per employed member). Two years of education seemed a very low threshold, so five years were used instead.[16] Also, given that the income indicator is being included, the high dependency index seemed less relevant in this hybrid approach. Finally, there are three indicators related to the dwelling's conditions. The first two, having proper sanitation (flush toilet) and living in a shelter with nonprecarious wall materials are typically included in the UBN approach.[17] The third indicator is having access to running water in the dwelling. Although this is not usually included in the UBN approach, it is considered important. In the absence of comparable health data, it can also be seen as a proxy of this dimension, which is one of the most valued according to the participatory study performed in Mexico

"Lo que dicen los pobres" (Székely, 2003). The overcrowding indicator typically included in the UBN approach (more than three people per room) could not be incorporated into the analysis because it was not present in some databases in some years.

In this chapter, all results correspond to estimates performed using equal weights for all the considered indicators, which does not mean that the *dimensions* are equally weighted. In fact, three of the indicators used refer to dwelling's characteristics and two other indicators (children attending school and the education of the household head) refer to the dimension of education of the household. Therefore, the equal weights are implicitly weighting the dwelling conditions three times, and the education dimension twice, compared to the income dimension. In Battistón, Cruces, Lopez-Calva, Lugo, and Santos (2009), we performed estimations using an alternative weighting system derived from a replica performed in Mexico – the participatory study "Voices of the poor" – carried out by the country's Secretaría de Desarrollo Social (Székely, 2003). In this scheme, the income dimension receives the highest weight, being 1.3 times the weight received by the children's education, 4 times the weight received by the education of the household head and access to running water, and 8 times the weight received by having access to sanitation and proper shelter. When appropriate, we mention the differences in results obtained with this alternative weighting structure.

4. RESULTS

This section presents the estimation of poverty figures for the six countries, using the multidimensional poverty measure presented in the previous section (M_0) for different values of the second cut-off $k = 1, 2, \ldots 6$, and using equal weights for all indicators.[18] Given that in two of the countries (Argentina and Uruguay) data for rural areas is not available, we have maintained urban and rural areas separately for all countries. For reference, the proportion of individuals living in urban and rural areas in each country can be found in Table A2. We highlight two results: the first, related to the evolution of multidimensional poverty, the second regarding the urban–rural disparities.

4.1. Evolution of Multidimensional Poverty

Fig. 1 presents the evolution of the adjusted headcount ratio M_0 for rural and urban areas, in panels A and B, respectively. The number of deprivation

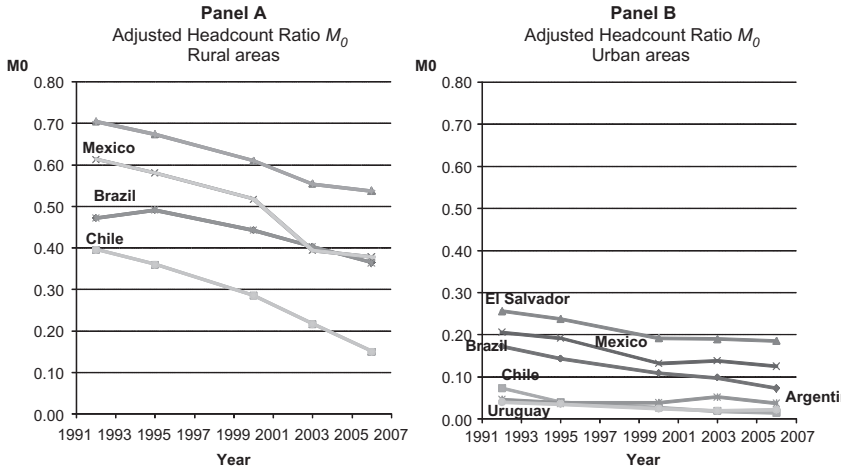

Fig. 1. Evolution over Time of M_0 with $k = 2$ and Equal Weights. *Source:* Authors' calculation, based on SEDLAC database.

used as the second cut-off is two. In other words, a person is considered to be poor if she falls short of the adequate level in two or more dimensions. $k = 2$ is chosen because it is the minimum k that requires an individual to be deprived in more than one indicator so as to be considered poor (i.e., it is "truly" *multidimensional*) and at the same time it is meaningful for all countries (for higher k values the aggregate M_0 estimate becomes virtually zero in the urban areas of Chile, Argentina, and Uruguay). Table A3 includes the estimation of M_0 for all possible values of k.[19]

A number of noteworthy points emerge from these graphs. First, in all countries considered multidimensional poverty decreased between 1992 and 2006. Although not presented here, this result is robust to the number of deprivations used as cut-offs. In most cases, the decrease was sizeable and uninterrupted throughout the period. In other cases, such as Argentina and Uruguay urban areas, multidimensional poverty either did not change significantly or was reduced only marginally.[20]

Second, multidimensional poverty in rural areas is significantly higher than in urban areas, especially at the beginning of the period. We come back to this point later in the section. Thirdly, and focusing on urban areas solely, one could categorize the countries in two groups: the countries belonging to the "Southern Cone" (Argentina, Chile, and Uruguay), with multidimensional M_0 estimates below 0.10, on the one hand, and Brazil, El Salvador,

and Mexico with poverty measurements closer to 0.20.[21] Finally, while the distinction between these two groups of countries is still apparent at the end of the period, there seems to be some convergence as the countries in the second group present a higher rate of reduction in multidimensional poverty than that of the first group.

As explained in Section 2.1, the M_0 measure is the product of two informative measures: the multidimensional headcount ratio H and the average deprivation share across the poor A. In order to better understand the drop in multidimensional poverty described in the previous paragraphs, we present in Fig. 2 the evolution of its two components (H and A) between 1992 and 2006, again for rural areas in panel A (excluding Argentina and Uruguay) and for urban areas in panel B. As before, k is set to two and all indicators are weighted "equally."

In both urban and rural areas for Brazil, Chile, El Salvador, and Mexico the reduction in multidimensional poverty M_0 is the result not only of the fall in the proportion of people deprived in two or more dimensions (H) but also of the fact that, on average, they became poor in fewer dimensions (A). However, the contribution of each of these components to the overall reduction of M_0 differs by country and area. For instance, in rural areas of Chile and urban areas of Brazil, most of the reduction seems to be driven by a substantial decline in the proportion of the population classified as poor, more than by a decrease in the average number of deprivations (even though, these all fell). On the contrary, in both the rural and urban areas of El Salvador the proportional reduction in A is larger than that of H, whereas in rural Mexico, both the percentage of the poor and the average deprivation seem to be reduced in similar proportions. Finally, in Uruguay and Argentina there was no significant reduction in the average number of deprivations experienced by the poor across the period. The only significant change is increase in the proportion of poor individuals in 2003 in Argentina, which is reflecting the increase in income poverty after the 2001 crisis.[22]

A second way of disentangling the meaning of the observed decrease in multidimensional poverty is to look at the evolution of the relative contribution of each of the dimensions into the total deprivation. As explained in Section 2.1 one of the advantages of the multidimensional poverty indices proposed by Alkire and Foster is that they can be broken-down by indicators. Fig. 3 presents the decomposition of M_0 with $k = 2$ and equal weights, panel A includes rural areas and panel B, urban areas. Table A4 presents the values for these decompositions.

Before looking at the evolution, it is worth examining which are the main contributors to total multidimensional poverty at the beginning of the

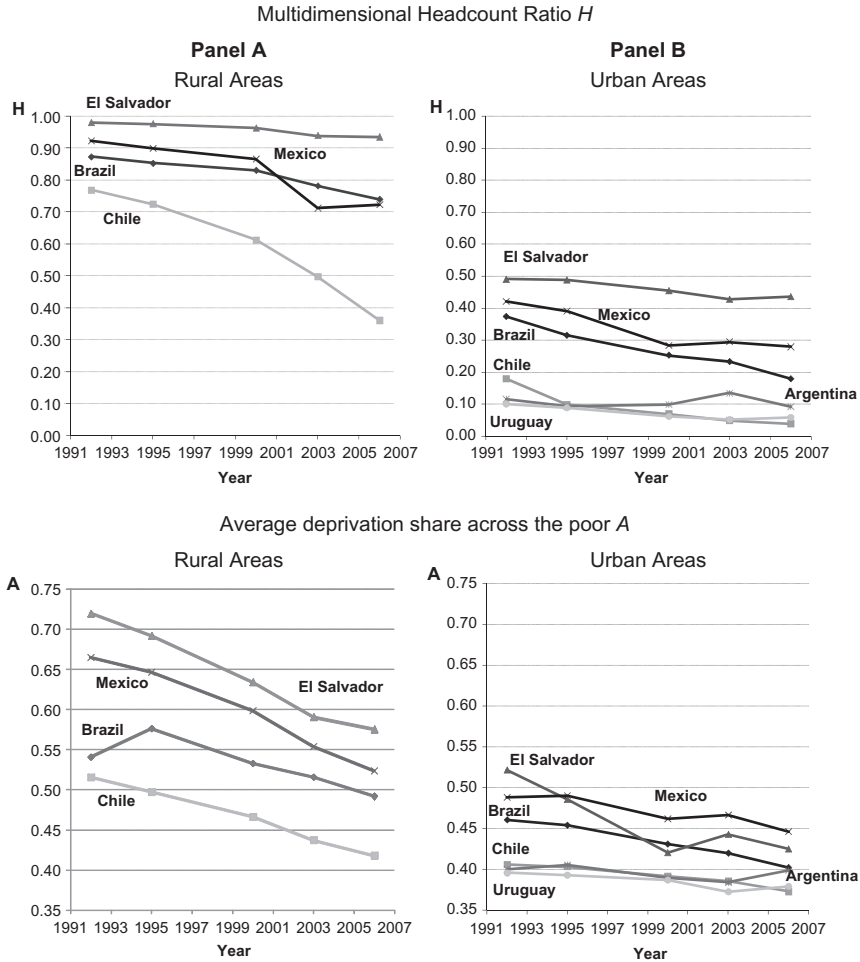

Fig. 2. Evolution over Time of Components of M_0 with $k = 2$ and Equal Weights. *Source:* Authors' calculation, based on SEDLAC database.

period. While the composition differs slightly across countries and region, attendance of children to school (second bar from the bottom) contributes relatively little to aggregate poverty, not exceeding 10 percent in most cases. On the contrary, insufficient education of the household head and poor sanitation appear in almost all countries and regions as the main contributors to poverty, the first contributing between 19 and 32 percent

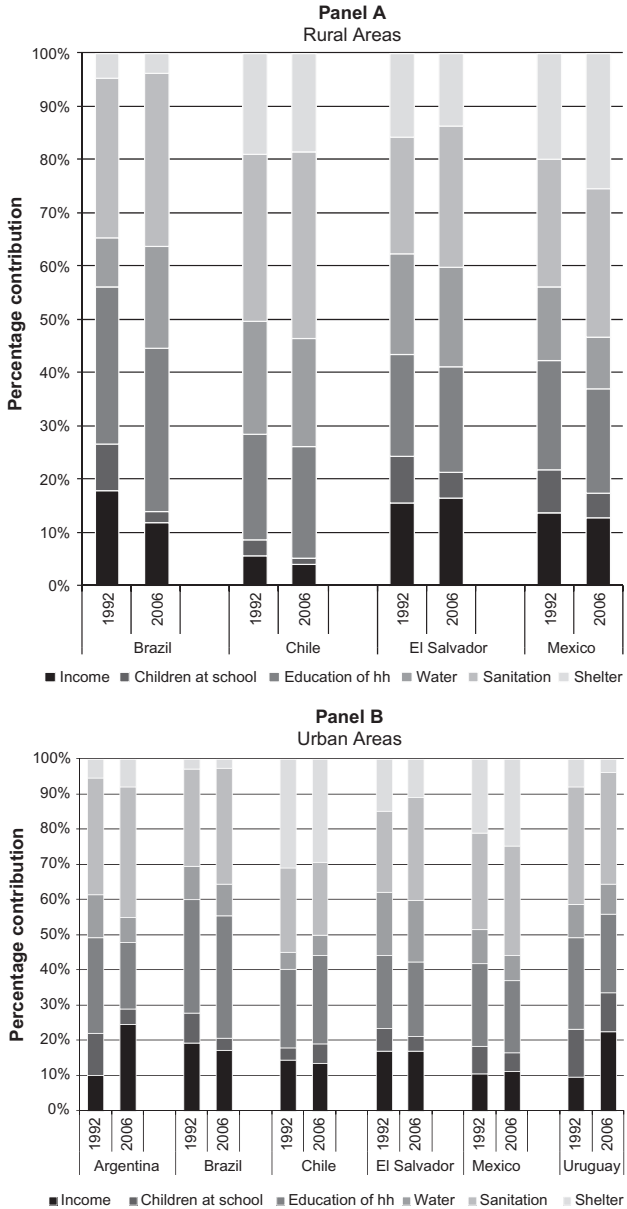

Fig. 3. Contribution of Deprivation in Each Indicator to Overall M_0 with $k = 2$ and Equal Weights. *Source:* Authors' calculation, based on SEDLAC database.

and the second contributing with 22 to 33 percent.[23] Two exceptions are urban areas in Chile and Mexico, where access to adequate shelter also explains a sizeable proportion of total poverty (31 and 21 percent, respectively). Finally, as expected, in rural areas the deprivations seem to be more concentrated in the dimensions related to infrastructure such as water, shelter, and sanitation. For instance, in Chile the contribution of deprivation in sanitation and running water is about one and a half and three and a half times that in urban areas. Similarly, in Brazil the contribution of water is more than twice that of urban areas.

In terms of the evolution, two points are worth highlighting. First, in Argentina and Uruguay urban areas – where multidimensional poverty has not decreased as much as in the other countries – there is an increase in the contribution of income to aggregate poverty. In both countries the contribution of income to total poverty soared from 10 percent to about 24 percent. This is consistent with the idea that between 1992 and 2006 there was a shift from "structural poor" to "new poor." Second, in the rest of the countries, most of the contributions of dimensions remain relatively constant. This, together with the previously observed reduction in aggregate poverty implies that there was a similar improvement in all dimensions of well-being included. Some exceptions include sanitation in urban areas of El Salvador and shelter in rural parts of Mexico that have seen their share rise considerably, from 23 to 29 percent and from 20 to 25 percent, respectively (Table A5).

4.2. Persistence of Poverty in Rural Areas

The second result of the present analysis regards the disparities between the urban and rural areas. The analysis is, therefore, restricted to the four countries where the information on both areas is available, that is, Brazil, Chile, El Salvador, and Mexico.

Despite the progress experienced in the countries of the region, rural areas still present high levels of multidimensional poverty. People living outside the cities are not only more likely to be multidimensionally poor than those in urban areas but also they are prone to experience multiple deprivations at the same time. This means that someone who falls short in one indicator of well-being (for instance, education) is quite likely to fall short also in another indicator (such as dwelling characteristics).

Fig. 4 presents for each country, disaggregated by urban and rural areas, the percentage of individuals deprived in one or more dimensions ($k = 1$),

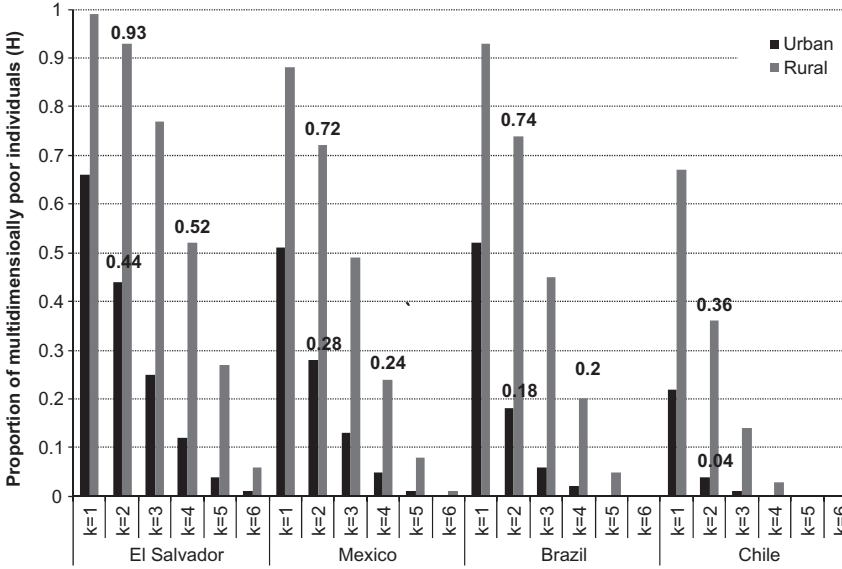

Fig. 4. Proportion of Individuals Deprived in $k = 1, 2 \ldots 6$ Dimensions and Equal Weights (Multidimensional Headcount H) – Year 2006. *Source:* Authors' calculation, based on SEDLAC database.

two or more ($k = 2$), and so on at the end of the period of study. Table A6 present the values of H for each country and area for the year 2006. In all four countries, the measurement of poverty is higher in rural areas than in its respective urban area for all possible k chosen, most often at least twice as large. Let us focus for the moment on multidimensional poverty as defined as those who are deprived in two or more indicator. While the proportion of poor in urban areas of El Salvador is 44 percent, more than twice (93 percent) are poor in rural areas. Similarly, in Mexico 28 percent of the urban population is poor whereas 72 percent of the rural one. In Brazil, the percentages are 18 and 74, respectively, while in Chile are 4 and 36 percent, respectively.

In addition, with the exception of Chile, the proportion of individuals suffering multiple deprivations is still significantly high in rural (above 20 percent) even when the required number of deprivations is four or more – out of six. At the extreme, we find El Salvador and Mexico, where the estimate of H falls below 5 percent only with the intersection approach, that is, when $k = 6$. These results point to a pattern in which people living in rural areas of Latin American who are deprived in one indicator are more

likely to fall below the minimum required in several other dimensions. This is also true in urban areas of El Salvador and, to a lesser extent, in urban Mexico. In contrast, people living in cities in Chile or Brazil who are deprived in an indicator are more likely to be deprived only in that single indicator (similarly for urban Argentina and Uruguay).

5. CONCLUSIONS

This chapter provides an analysis of multidimensional poverty in Argentina, Brazil, Chile, El Salvador, Mexico, and Uruguay for the period between 1992 and 2006. We use an approach that intends to improve the UBN method used in the region in three ways: first, using a hybrid approach in the selection of indicators incorporated in the multidimensional measure, including variables normally used by both the UBN and those who prefer the income method; second, by using one of the measures of the family of multidimensional poverty indices proposed by Alkire and Foster (2007), we can account for both poverty incidence and the average proportion of deprivations that the poor experience. The index also allows alternative weighting schemes, although in this chapter we weigh all indicators equally.

The overall picture from the present analysis is mixed. On the one hand, there has been an enormous improvement in all 6 countries and regions in the 14 years considered, with decreasing trends in multidimensional poverty due, in general, to both a reduction in the proportion of individuals that are poor and the number of deprivations that they have on average. With the exception of urban Argentina and Uruguay, the overall reduction seems to come from an equal improvement in all indicators. Instead, in Argentina and Uruguay, the increase in the importance of income poverty in the overall multidimensional poverty reflects the increase in deprivation in this dimension over the period.

The second result of the chapter relates to the fact that multidimensional poverty estimates in rural areas are still considerably high. Even when we observe a trend toward convergence with urban areas rates, there seems to be a pattern: when people living outside the cities do not reach the adequate level of achievement in a given indicator, they are more likely to be deprived in several other indicators as well. In contrast, inhabitants of Chilean and Brazilian cities who fall short in a given indicator of well-being, they tend to be deprived only in that single indicator. This is true to a lesser extent in urban areas of Mexico and El Salvador. Such results suggest the need of a renewed effort to tackle poverty in rural areas.

The methodological improvements proposed in this chapter highlight the need for a thorough (public) discussion on the full set of dimensions, indicators, and weights to be included in a multidimensional poverty measure for the region. This discussion could contribute to the development of a tool that becomes relevant, meaningful, and useful to track progress in the Millennium Development Goals as well as in social policy in general. Such index will, most likely, demand improving the collection of information in several key dimensions of well-being.

UNCITED REFERENCES

Beccaria & Minujin (1988); CEPAL and Direccion General de Estadistica y Censos del Uruguay (1988); Ravallion, Chen, & Sangrula (2008).

NOTES

1. Household surveys did not become regular until the 1970s or even later in Latin American countries and even when they were performed, microdatasets were not publicly available for researchers (Gasparini, 2004).

2. Cruces and Gasparini (2008) illustrate these inclusion and exclusion effects by studying the targeting of cash transfer programs based on a combination of income and other UBN-related indicators.

3. Note that this is also the procedure followed when constructing the Human Development Index.

4. In the case of the six countries considered in this study, we found that the Spearman correlation of income with the other considered dimension does not exceed 0.5 in any case and it is decreasing over time. See these results in Battistón et al. (2009).

5. Several other unidimensional poverty measures have been proposed that satisfy monotonicity and transfer: the mentioned papers of Watts and Sen propose measures themselves, as well as Foster et al. (1984), Clark, Hemming, and Ulph (1981), Chakravarty (1983), among others. Foster (2006), Foster and Sen (1997), and Atkinson (1987) provide excellent surveys of unidimensional poverty measures and the desirable axiomatic framework.

6. The family is defined as $M_\alpha(X;z) = (1/nd)\sum_{i=1}^{n}\sum_{j=1}^{d} w_j(g_{ij}(k))^\alpha$ with $\alpha \geq 0$, where $g_{ij}(k)$ is the censored poverty gap of individual i in dimension j: $g_{ij}(k) = \lfloor (z_j - x_{ij})/z_j \rfloor$, if $x_{ij} < z_j$ and $c_i \geq k$, and $g_{ij}(k) = 0$ otherwise; w_j is the weight assigned to dimension j, such that $\sum_{j=1}^{d} w_j = d$, and α is the parameter of dimension-specific poverty aversion. The M_α family is presented in Alkire and Foster (2007) as the mean of the censored matrix of normalized alpha poverty gaps.

7. In Battistón et al. (2009) we present results for other members of the Alkire and Foster family of measures which account for depth and severity (namely M_1 and

M_2). In that paper, we acknowledge the technical problems of using such measures when not all the indicators are cardinal.

8. Given a population subgroup I, its contribution to overall poverty is given by $C_I = [(n_I/n)M_\alpha^I]/M_\alpha$.

9. Specifically, the contribution of dimension J is given by $C_J = [((1/nd)\sum_{i=1}^{n} (g_{iJ}(k))^\alpha)]/M_\alpha$.

10. On the meaning of dimension weights in multidimensional indices of well-being and deprivation and alternative approaches to setting them, see Decanq and Lugo (2009).

11. The SEDLAC database (CEDLAS & World Bank, 2009) will report multidimensional poverty indicators systematically starting in 2009.

12. In Chile, it corresponds to localities of less than 1,000 people or with 1,000–2,000 people, of which most perform primary activities. In Mexico, it refers to localities of less than 2,500 people. In Brazil, rural areas are not defined according to population size but rather they are all those not defined as urban agglomerations by the Brazilian Institute of Geography and Statistics. In El Salvador, rural areas are all those outside the limits of municipalities heads, which are populated centers where the administration of the municipality is located. Again, this definition does not refer to any particular population size.

13. The Statistics Institute of each country has a criterion to identify invalid income answers (such as reporting zero income when working for a salary), which is incorporated in the SEDLAC dataset, as well as other types of invalid answers (such as reporting labor income when being unemployed).

14. The dollar-a-day lines are not without problems. It can be argued, for instance, that the national poverty lines computed by each country's statistical institute tend to be more accurate and appropriate than the World Bank's ones as the former are based on nationally relevant food baskets (Reddy & Pogge, 2010). Still, we use the World Bank lines for this study because they stem from a common methodology and provide benchmarks for international comparisons. Moreover, these are the lines considered by the monitoring indicators of the Millennium Development Goals. See Reddy and Pogge (2010) for critique and alternative approach, and Ravallion (2010) for a response. See also Deaton's (2010) presidential address to the American Economic Association on the difficulties of dealing with PPP values for international comparisons of poverty.

15. In Brazil, household income is expressed in gross terms (rather than net terms, as in other countries). Fortunately, during the period under analysis there have not been major changes in the labor or income tax, hence the comparability across years is not compromised.

16. Five years of complete schooling is seen here as a good "compromise," albeit arbitrary, between alternative thresholds used by countries in computing UBN and current demands of the labor market. This is set as a (lower bound) approximation to complete primary education. UNESCO considers that in Argentina, Mexico, El Salvador, Chile and Uruguay, the equivalent to the first level of the International Standard Classification of Education (1997) lasts 6 years, while it lasts 4 in Brazil.

17. This definition of the shelter indicator in this paper is slightly different than the one used by the UBN approach, which uses "adequate shelter." Only in the case of Uruguay, we used "adequate shelter," as the question on wall materials was not included

after the year 2000. The SEDLAC database defines precarious house from Uruguayan EPH as a room inside a house, fully occupied as a residence by a single household.

18. As explained in Section 2.1, Battistón et al. (2009) also compute all combinations of multidimensional poverty using "voices of the poor" weights. The main conclusions of this section are consistent with the results using this alternative weighting system, even though, as expected, the estimates follow more closely the evolution of the higher-weighted dimensions, that is,. income and education. For the complete estimation results see Battistón et al. (2009).

19. Note that with $k = 2$ someone deprived only in income (or in any other indicator) is not counted as poor. We purposely choose this approach since we are aiming at identifying people with *coupled* deprivations, that is, more than one. However, Tables A3 and A6 provide the M_0 and H estimates for all possible k values.

20. To unambiguously assess the reductions in poverty, all estimates were bootstrapped using 200 replications. Results of the bootstraps can be found in a companion document to Battistón et al. (2009) "WP 17 Bootstrapped Estimates and Correlations" (http://www.ophi.org.uk/subindex.php?id = publications0).

21. One should remember that these figures cannot be interpreted as percentage of population that are multidimensionally poor but rather as the combination of that proportion and the average share of deprivation suffered by this population. Therefore, a value of 0.10 could be the result 30 percent of individuals classified as poor with, on average, two (out of six) deprivations, or 15 percent of individuals with an average of four (out of six) deprivations.

22. See also Table A5, which reports the headcount ratios by indicator.

23. The high contribution of sanitation can be due to the threshold used: we require flush toilet to be nondeprived.

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APPENDIX

Table A1. Sample Size for each Country and Year, Rural, and Urban Areas.

Country	Household Survey	Year	Sample Size (People)	
			Urban	Rural
Argentina[a]	Encuesta Permanente de Hogares (EPH)	1992	59,528	NA
		1995	62,372	NA
		2000	43,255	NA
		2003	29,075	NA
	Encuesta Permanente de Hogares Continua (EPH-C)	2006	45,676	NA
Brazil	Pesquisa Nacional por Amostra de Domicilios (PNAD)	1992	244,473	55,544
		1995	266,287	57,859
		2001	316,860	52,753
		2003	322,839	53,932
		2006	337,509	65,372
Chile	Encuesta de Caracterizacion Socioeconomica Nacional (CASEN)	1992	86,179	46,698
		1996	94,925	32,500
		2000	142,029	89,441
		2003	150,156	80,411
		2005	153,234	86,058
El Salvador	Encuesta de Hogares de Propositos Multiples (EHPM)	1991	49,243	39,235
		1995	20,989	18,009
		2000	40,940	29,843
		2003	35,622	35,708
		2005	34,127	35,517
Mexico	Encuesta Nacional de Ingresos y Gastos de los Hogares (ENIGH)	1992	27,913	20,265
		1996	39,974	21,840
		2000	26,402	13,989
		2004	68,016	21,907
		2006	58,760	23,140
Uruguay	Encuesta Continua de Hogares (ECH)	1992	28,658	NA
		1995	64,177	NA
		2000	51,913	NA
		2003	54,750	NA
		2005	53,738	NA

[a]For the sake of comparability over time, the samples used correspond to the same 15 urban agglomerations.

Table A2. Percentage of Urban Population.

Year	Brazil	Chile	El Salvador	Mexico
1992	0.8	0.84	0.52	0.76
1995	0.81	0.85	0.59	0.77
2001	0.85	0.86	0.63	0.78
2003	0.86	0.87	0.62	0.77
2006	0.85	0.87	0.63	0.78

Source: Authors' calculations, based on SEDLAC database.

Table A3. Multidimensional Poverty M_0 for Alternative k and Equal Weights.

Country	Year	Rural						Urban					
		$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
Argentina	1992							0.080	0.046	0.021	0.005	0.001	0.000
	1995							0.070	0.038	0.017	0.006	0.000	0.000
	2000							0.071	0.039	0.015	0.003	0.000	0.000
	2003							0.088	0.052	0.019	0.003	0.001	0.000
	2006							0.070	0.037	0.016	0.005	0.001	0.000
Brazil	1992	0.487	0.472	0.388	0.245	0.093	0.018	0.227	0.173	0.108	0.056	0.020	0.004
	1995	0.509	0.491	0.424	0.298	0.138	0.031	0.200	0.143	0.087	0.044	0.016	0.003
	2000	0.463	0.442	0.357	0.229	0.075	0.008	0.166	0.109	0.059	0.026	0.007	0.001
	2003	0.429	0.402	0.313	0.190	0.056	0.004	0.154	0.098	0.049	0.020	0.004	0.000
	2006	0.395	0.364	0.267	0.145	0.039	0.004	0.129	0.072	0.031	0.011	0.002	0.000
Chile	1992	0.421	0.396	0.319	0.170	0.049	0.006	0.124	0.073	0.033	0.011	0.003	0.000
	1995	0.389	0.360	0.273	0.140	0.037	0.006	0.073	0.040	0.018	0.005	0.001	0.000
	2000	0.324	0.285	0.196	0.085	0.017	0.003	0.059	0.027	0.010	0.003	0.001	0.000
	2003	0.263	0.217	0.130	0.045	0.007	0.000	0.047	0.018	0.007	0.002	0.000	0.000
	2006	0.202	0.151	0.079	0.023	0.004	0.000	0.044	0.014	0.004	0.001	0.000	0.000
El Salvador	1992	0.732	0.729	0.711	0.641	0.463	0.180	0.311	0.278	0.230	0.166	0.086	0.024
	1995	0.706	0.703	0.682	0.600	0.401	0.153	0.297	0.264	0.213	0.137	0.065	0.019
	2000	0.654	0.648	0.613	0.513	0.314	0.095	0.267	0.231	0.171	0.105	0.047	0.007
	2003	0.605	0.597	0.549	0.435	0.246	0.065	0.252	0.213	0.154	0.092	0.037	0.005
	2006	0.590	0.582	0.527	0.403	0.234	0.057	0.251	0.214	0.153	0.085	0.034	0.007
Mexico	1992	0.622	0.613	0.573	0.476	0.291	0.096	0.243	0.206	0.145	0.080	0.033	0.007
	1995	0.592	0.581	0.539	0.426	0.246	0.084	0.230	0.192	0.137	0.077	0.027	0.004
	2000	0.532	0.517	0.464	0.338	0.161	0.030	0.174	0.131	0.083	0.043	0.015	0.003
	2003	0.427	0.394	0.325	0.224	0.100	0.016	0.180	0.137	0.091	0.044	0.014	0.003
	2006	0.405	0.378	0.300	0.178	0.068	0.013	0.164	0.125	0.075	0.034	0.010	0.001
Uruguay	1992							0.078	0.040	0.016	0.005	0.001	0.000
	1995							0.073	0.035	0.014	0.004	0.000	0.000
	2000							0.056	0.024	0.008	0.002	0.000	0.000
	2003							0.050	0.019	0.006	0.002	0.000	0.000
	2006							0.053	0.022	0.009	0.001	0.000	0.000

Source: Authors' calculations, based on SEDLAC database.

Table A4. Decomposition of Multidimensional Poverty M_0 for $k = 2$ and Equal Weights.

Country	Year	Rural						Urban					
		Income	Children at school	Education of household head	Water	Sanitation	Shelter	Income	Children at school	Education of household head	Water	Sanitation	Shelter
Argentina	1992							0.103	0.118	0.272	0.122	0.332	0.053
	2006							0.248	0.044	0.189	0.069	0.372	0.079
Brazil	1992	0.181	0.086	0.295	0.091	0.299	0.048	0.195	0.084	0.322	0.095	0.276	0.029
	2006	0.121	0.021	0.305	0.190	0.324	0.039	0.174	0.033	0.349	0.089	0.328	0.027
Chile	1992	0.057	0.031	0.199	0.211	0.311	0.191	0.146	0.035	0.222	0.049	0.238	0.310
	2006	0.041	0.012	0.209	0.204	0.349	0.185	0.137	0.055	0.252	0.056	0.207	0.294
El Salvador	1992	0.156	0.088	0.192	0.189	0.218	0.157	0.170	0.065	0.207	0.180	0.230	0.149
	2006	0.167	0.048	0.198	0.186	0.265	0.136	0.170	0.043	0.212	0.175	0.292	0.109
Mexico	1992	0.140	0.078	0.205	0.137	0.240	0.199	0.106	0.079	0.235	0.097	0.272	0.212
	2006	0.129	0.046	0.196	0.098	0.278	0.254	0.114	0.052	0.204	0.073	0.311	0.247
Uruguay	1992							0.097	0.137	0.259	0.094	0.335	0.079
	2006							0.225	0.111	0.224	0.085	0.318	0.037

Source: Authors' calculations, based on SEDLAC database.

Country	Year	Rural						Urban					
		Income	Children at school	Education of household head	Water	Sanitation	Shelter	Income	Children at school	Education of household head	Water	Sanitation	Shelter
Argentina	1992							0.042	0.057	0.174	0.032	0.143	0.019
	1995							0.075	0.047	0.126	0.031	0.111	0.020
	2000							0.094	0.012	0.121	0.016	0.158	0.013
	2003							0.221	0.010	0.106	0.012	0.154	0.012
	2006							0.091	0.017	0.096	0.017	0.176	0.024
Brazil	1992	0.514	0.253	0.885	0.272	0.898	0.162	0.233	0.094	0.552	0.100	0.341	0.031
	1995	0.393	0.189	0.863	0.606	0.870	0.155	0.145	0.071	0.527	0.114	0.317	0.028
	2000	0.390	0.062	0.838	0.512	0.863	0.122	0.165	0.029	0.446	0.075	0.261	0.017
	2003	0.352	0.053	0.796	0.452	0.829	0.095	0.166	0.023	0.414	0.061	0.244	0.014
	2006	0.270	0.046	0.750	0.419	0.798	0.085	0.106	0.018	0.368	0.041	0.223	0.013
Chile	1992	0.140	0.075	0.504	0.505	0.818	0.485	0.096	0.021	0.198	0.022	0.136	0.270
	1995	0.136	0.054	0.483	0.499	0.796	0.395	0.056	0.018	0.154	0.013	0.090	0.108
	2000	0.093	0.035	0.436	0.412	0.688	0.321	0.053	0.015	0.138	0.010	0.061	0.081
	2003	0.080	0.020	0.329	0.307	0.592	0.275	0.046	0.010	0.104	0.007	0.049	0.069
	2006	0.045	0.014	0.282	0.226	0.434	0.233	0.030	0.011	0.113	0.007	0.030	0.074
El Salvador	1992	0.683	0.385	0.844	0.829	0.964	0.688	0.321	0.122	0.431	0.307	0.407	0.281
	1995	0.612	0.303	0.822	0.826	0.974	0.696	0.257	0.098	0.430	0.340	0.404	0.251
	2000	0.640	0.233	0.774	0.698	0.980	0.594	0.221	0.068	0.369	0.253	0.487	0.197
	2003	0.603	0.192	0.720	0.649	0.962	0.507	0.267	0.062	0.332	0.254	0.451	0.145
	2006	0.584	0.168	0.697	0.654	0.962	0.476	0.256	0.065	0.347	0.239	0.448	0.153
Mexico	1992	0.515	0.290	0.771	0.506	0.907	0.738	0.146	0.108	0.388	0.121	0.383	0.313
	1995	0.650	0.240	0.686	0.407	0.887	0.674	0.286	0.083	0.307	0.078	0.369	0.259
	2000	0.537	0.169	0.658	0.263	0.849	0.721	0.128	0.060	0.257	0.046	0.300	0.253
	2003	0.369	0.097	0.558	0.275	0.661	0.628	0.118	0.059	0.255	0.078	0.290	0.281
	2006	0.301	0.111	0.495	0.234	0.684	0.609	0.103	0.050	0.222	0.060	0.303	0.251
Uruguay	1992							0.029	0.050	0.203	0.028	0.123	0.032
	1995							0.029	0.042	0.213	0.025	0.105	0.024
	2000							0.027	0.037	0.145	0.026	0.081	0.018
	2003							0.049	0.031	0.131	0.019	0.063	0.010
	2006							0.060	0.029	0.127	0.017	0.068	0.014

Source: Authors' calculations, based on SEDLAC database.

Table A6. Multidimensional Headcount Rates with Equal Weights (Year 2006).

Country	Rural						Urban					
	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
Argentina							0.29	0.10	0.03	0.01	0.00	0.00
Brazil	0.93	0.74	0.45	0.20	0.05	0.00	0.52	0.18	0.06	0.02	0.00	0.00
Chile	0.67	0.36	0.14	0.03	0.00	0.00	0.22	0.04	0.01	0.00	0.00	0.00
El Salvador	0.99	0.93	0.77	0.52	0.27	0.06	0.66	0.44	0.25	0.12	0.04	0.01
Mexico	0.88	0.72	0.49	0.24	0.08	0.01	0.51	0.28	0.13	0.05	0.01	0.00
Uruguay							0.24	0.06	0.02	0.00	0.00	0.00

Source: Authors' calculations, based on SEDLAC database.